

Mueller matrix ellipsometry with imperfect compensators

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We describe the use of an ellipsometer having the configuration PCSC'A to determine the Mueller matrix of a linear optical system S, using imperfect compensators C and C' and linear polarizer P and analyzer A. Two manual (16 intensity), one semiautomatic (rotating-compensator) and one fully automatic (dual rotating-compensator) methods are described. Real-time Fourier analysis (of both raw and reduced data) is used in the automatic methods to reduce random and systematic measurement errors. Calibration procedures given for each method permit Mueller matrix *spectroscopy* using (quasi-) achromatic retarders whose principal axes, as well as retardation and transmission ratios change slightly with wavelength.

I. INTRODUCTION

Broadly speaking, ellipsometry¹⁻⁴ concerns the measurement of the polarization-altering properties of optical systems. Recent advances in spectroscopic ellipsometry⁵ have increased the power and usefulness of this technique. Ellipsometry is usually applied to nondepolarizing systems, providing a partial or complete determination of the system Jones matrix.⁶⁻⁸ These systems can be measured by null methods⁹ [e.g., with a classical polarizer-compensator-sample-analyzer (PCSA) ellipsometer], since totally polarized light can be extinguished with a retarder and analyzer. In contrast, the Mueller matrix⁶ of a depolarizing system cannot be measured by null methods, and so classical ellipsometry (i.e., null ellipsometry) cannot be applied to this problem.

Photometric ellipsometry¹⁰ (or photopolarimetry)^{11,12} combines photometry with the polarimetric techniques of ellipsometry to permit measurement of all four Stokes parameters of a beam of light. Thus photopolarimetry, in any of several forms, can provide a measure of the degree of polarization of light as well as the state of polarization of the polarized component.¹³ It is possible, then, to apply photometric ellipsometry to the characterization of depolarizing optical systems.¹⁴ The term Mueller-matrix ellipsometry (MME) in this paper denotes photopolarimetric measurement of the Mueller matrix of a linear depolarizing optical system.

A Mueller-matrix ellipsometer, as indicated in Fig. 1, provides input light of known polarization states via a polarization-state generator (PSG), permits interaction of the light with the optical system being measured, and provides a measurement of the output (scattered) polarization states via a polarization-state detector (PSD) or polarimeter. An appropriate set of measurements is then analyzed to obtain the Mueller matrix.

Depolarizing systems generally exhibit particulate scattering^{15,16} (aerosols, colloidal suspensions, opalescent materials), rather than specular reflection or refraction. The Mueller matrix is investigated, usually as a function of wavelength and scattering angle, in order to determine the shape, refractive index, mean size, and spread of the size distribution of the scattering particles.¹⁷

We do not discuss all the possible methods for performing MME. We do note here that polarization-modulation techniques,¹⁸ which permit very short measurement times, have recently been adapted for MME in both semi-automatic¹⁹ and fully-automatic²⁰ configurations. Multiple phase-sensitive

detectors are required if the high measurement speed is to be fully exploited, however.²⁰ Other photopolarimetric methods can also be used for MME.

This paper describes the PCSC'A ellipsometer, derived from the classical PCSA instrument by the addition of a compensator C' in the output path, and its application to MME by four different methods, with various degrees of automation. In automatic (i.e., rotating-compensator) modes of operation, redundant measurements reduce random and systematic errors and provide estimates of their magnitude.

The next section gives an analysis of the PCSC'A ellipsometer using the Mueller-Stokes calculus. It contains a derivation of the detector response in terms of the system Mueller matrix, the component azimuths, compensator defect parameters, and the response to the source intensity. A method of calibrating the instrument is included. These results form the basis for the description of four methods of MME.

II. THE PCSC'A ELLIPSOMETER

In the PCSC'A ellipsometer, shown in its essential form in Fig. 2, the source, polarizer P, and compensator C in the input arm serve as a polarization-state generator (PSG), while the compensator C', analyzer A, and linear detector in the output arm serve as the polarization-state detector (PSD). The components of the ellipsometer are arbitrary except as follows: P and A are perfect polarizers, while C and C' each possess a diagonal Jones matrix and a retardance other than multiple half wave (0°, 180°, ..., etc.). The source and detector characteristics are not specified other than that they be compatible with each other and with the polarizing elements used. The system Mueller matrix is arbitrary.

We follow the Muller-Nebraska convention²¹ throughout this paper. The plane of scattering serves as the 0° azimuth reference for the Stokes vectors and optical components. Rotations and azimuths are measured from the reference axis ccw looking into the source. By definition, the Mueller matrix has the same reference axes as the Stokes vectors it relates. It may be pre- and post-multiplied by appropriate rotation matrices subsequent to its measurement to obtain reference to any other desired axes.

A. Analysis

The Jones matrix of the compensator C when the fast-axis azimuth $C = 0^\circ$ is given by²²



FIG. 1. Necessary components of a Mueller matrix ellipsometer: PSG is a polarization-state generator, PSD is a polarization-state detector.

$$C_J = \begin{bmatrix} 1 & 0 \\ 0 & \rho_C \end{bmatrix} \quad (1)$$

where

$$\rho_C = \tan\psi_C e^{-i\delta_C} = T e^{-i\delta_C} \quad (2)$$

is the complex ratio of the slow-to-fast-axis transmission coefficients. [The trigonometric form given first in Eq. (2) has some notational advantages over the more common notation, which is also given here and in Eqs. (4) for reference.] The corresponding Mueller matrix for $C = 0^\circ$ is

$$C = \begin{bmatrix} 1 & s & 0 & 0 \\ s & 1 & 0 & 0 \\ 0 & 0 & p & r \\ 0 & 0 & -r & p \end{bmatrix} \quad (3)$$

The elements of C_J and C are related by

$$\begin{aligned} p &= \sin 2\psi_C \cos \delta_C = 2T \cos \delta_C / (1 + T^2) \cong 0, \\ r &= \sin 2\psi_C \sin \delta_C = 2T \sin \delta_C / (1 + T^2) \cong 1, \\ s &= \cos 2\psi_C = (1 - T^2) / (1 + T^2) \cong 0. \end{aligned} \quad (4)$$

The approximations indicated in Eqs. (4) are strict equalities for an ideal quarter-wave retarder ($T = 1$, $\psi_C = 45^\circ$, $\delta_C = 90^\circ$). (Note that p , r , and s are not mutually independent, since their squares sum identically to unity.)²³

For an arbitrary orientation of the compensator fast axis $C \cong 0^\circ$, C becomes

$$C = \begin{bmatrix} 1 & a & b & 0 \\ a & u & v & -x \\ b & v & w & y \\ 0 & x & -y & p \end{bmatrix}, \quad (5)$$

where

$$\begin{aligned} a &= s \cos 2C, & u &= f \cos 4C + (1 - f), \\ b &= s \sin 2C, & v &= f \sin 4C, \\ x &= r \sin 2C, & w &= -f \cos 4C + (1 - f), \\ y &= r \cos 2C, & f &= (1 - p)/2 \simeq 0.5. \end{aligned} \quad (6)$$

A similar matrix C' describes the behavior of the compensator C' .

The Stokes vector S generated by the PSG is then given by

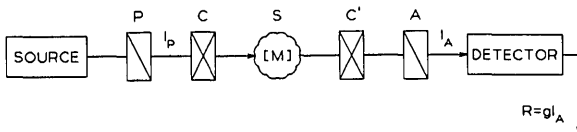


FIG. 2. PCSC'A ellipsometer. I_p and I_A are the intensities leaving the polarizer and analyzer, respectively. The compensators C and C' are imperfect (see text.) The detector is linear.

$$\begin{bmatrix} S_0 \\ S_1 \\ S_2 \\ S_3 \end{bmatrix} = \begin{bmatrix} 1 & a & b & 0 & 1 \\ a & u & v & -x & \cos 2P \\ b & v & w & y & \sin 2P \\ 0 & x & -y & p & 0 \end{bmatrix} I_P \quad (7)$$

$$= \begin{bmatrix} s \cos(2C - 2P) + 1 \\ f \cos(4C - 2P) + s \cos 2C + (1 - f) \cos 2P \\ f \sin(4C - 2P) + s \sin 2C + (1 - f) \sin 2P \\ r \sin(2C - 2P) \end{bmatrix} I_P, \quad (8)$$

where I_P is the intensity of the light leaving the polarizer, (independent of P if P must change,) and the column matrix in Eq. (7) is the first column in the Mueller matrix of the polarizer at azimuth P . The column vector in Eq. (8) contains coefficients of I_P which generate the input Stokes parameters.

The system Mueller matrix M transforms the input Stokes vector S into the output Stokes vector $S' = MS$ as follows:

$$\begin{bmatrix} S'_0 \\ S'_1 \\ S'_2 \\ S'_3 \end{bmatrix} = \begin{bmatrix} M_{00} & M_{01} & M_{02} & M_{03} \\ M_{10} & M_{11} & M_{12} & M_{13} \\ M_{20} & M_{21} & M_{22} & M_{23} \\ M_{30} & M_{31} & M_{32} & M_{33} \end{bmatrix} \begin{bmatrix} S_0 \\ S_1 \\ S_2 \\ S_3 \end{bmatrix}. \quad (9)$$

This is in turn transformed by the second compensator and analyzer so that the Stokes vector of the light leaving the analyzer (referenced to the transmitting axis of the analyzer) is given by

$$\begin{bmatrix} I_A \\ I_A \\ 0 \\ 0 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 & \cos 2A & \sin 2A & 0 \\ 1 & \cos 2A & \sin 2A & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \times \begin{bmatrix} 1 & a' & b' & 0 \\ a' & u' & v' & -x' \\ b' & v' & w' & y' \\ 0 & x' & -y' & p' \end{bmatrix} \begin{bmatrix} S'_0 \\ S'_1 \\ S'_2 \\ S'_3 \end{bmatrix} \quad (10)$$

From Eqs. (6) for C' , the intensity I_A leaving the analyzer and incident on the detector is

given by

$$\begin{aligned} 2I_A &= S'_0[s' \cos(2C' - 2A) + 1] \\ &+ S'_1[f' \cos(4C' - 2A) + s' \cos 2C' + (1 - f') \cos 2A] \\ &+ S'_2[f' \sin(4C' - 2A) + s' \sin 2C' + (1 - f') \sin 2A] \\ &+ S'_3[-r' \sin(2C' - 2A)]. \end{aligned} \quad (11)$$

There is a similarity of the coefficients in Eq. (11) to those in Eq. (8). They are the elements of the row vector formed by the product of the top row of the analyzer matrix in Eq. (10) and the following compensator matrix. Equation (11) is the mathematical dual of Eq. (8), containing the coefficients of the output Stokes parameters which generate the intensity falling on the detector. The factor 2 would be present in Eq. (8) if we considered I_P to be the (unpolarized) intensity incident on the polarizer.

Equations (7)–(11) can be written in matrix form by defining the column matrix

$$X = \{1, \cos 2X, \sin 2X, 0\}, \quad (12)$$

where $X = P(A)$ is the first column in the Mueller matrix of the polarizer (analyzer) set at an azimuth $X = P(A)$. Then

$$I_A = (1/2)(A^t C' M C P) I_P, \quad (13)$$

where t signifies matrix transpose. Equation (13) has the form $I_A = T_E I_P$, where T_E is a scalar and represents the intensity transmittance of the ellipsometer. In essence, MME consists of measuring T_E under an appropriate set of conditions, and inferring $\mathbf{M}$ from *a priori* knowledge of $\mathbf{A}$, $\mathbf{C}'$, $\mathbf{C}$, and $\mathbf{P}$ obtained from calibration.

Finally, the detector has a gain g , (independent of A if A must change,) so that the detector response R in arbitrary units is

$$R = g I_A \quad (14)$$

$$= (g I_P / 2) \mathbf{A}^t \mathbf{C}' \mathbf{M} \mathbf{C} \mathbf{P}. \quad (15)$$

This equation gives the measured response R for a single set of experimental conditions (e.g., component azimuth values). For a series of measurements, a matrix $\mathbf{R}$ of results can be expressed in a similar manner by defining the generator and detector matrices

$$\begin{aligned} \mathbf{G} &\equiv \mathbf{C} \mathbf{P}, \\ \mathbf{D} &\equiv \mathbf{A}^t \mathbf{C}', \end{aligned} \quad (16)$$

where $\mathbf{G}(\mathbf{D})$ contains a single column (row) for each pair of azimuth values $P, C(C', A)$ in the PSG (PSD). Equations (15) and (16) then define a response matrix

$$\mathbf{R} = (g I_P / 2) \mathbf{D} \mathbf{M} \mathbf{G}. \quad (17)$$

The elements R_{ij} of $\mathbf{R}$ correspond to the combination of the i th PSD and j th PSG settings.

If four azimuth settings for both the PSG and PSD are judiciously chosen, so that $\mathbf{D}$ and $\mathbf{G}$ both possess inverses, Eq. (17) yields a direct solution for $\mathbf{M}$:

$$\mathbf{M} = (2/g I_P) \mathbf{D}^{-1} \mathbf{R} \mathbf{G}^{-1}. \quad (18)$$

The use of Eq. (18) requires four input Stokes vectors that are linearly independent, and four output intensities which depend on the output Stokes parameters in a linearly independent manner. This type of determination of $\mathbf{M}$ is called a direct or 16-intensity method of MME.

B. Calibration

The ellipsometer may be calibrated manually if we assume azimuth scales for the compensators and analyzer, and the ability to distinguish between the compensator fast and slow axes. The arms are made collinear, and all components are removed. The polarizer $\mathbf{P}$ is fixed by definition at $P = 0^\circ$. The analyzer is inserted and adjusted for null so that $A = 90^\circ$. The compensator $\mathbf{C}$ is inserted and adjusted for the null at $C = 0^\circ$. The $\mathbf{C}$ and $\mathbf{A}$ scales are now calibrated. Next the analyzer is rotated 90° so that $A = 0^\circ$, and three intensity readings are taken for $C = 0^\circ, 45^\circ$, and 90° . From Eqs. (8), (9), (11), and (14), these three readings are, respectively,

$$\begin{aligned} R_1 &= (1 + s) g I_P, \\ R_2 &= (1 - f) g I_P, \\ R_3 &= (1 - s) g I_P. \end{aligned} \quad (19)$$

[Note that the absence of $\mathbf{C}'$ is equivalent to setting $s' = f' = r' = 0$ in Eq. (11), and the absence of the system $\mathbf{S}$ is equivalent to setting $\mathbf{M}$ equal to the identity matrix $\mathbf{I}$ in Eq. (9).] Now $\mathbf{C}$ is removed and $\mathbf{C}'$ is inserted. The $\mathbf{C}'$ azimuth scale is cal-

ibrated with a PC'A null, and three readings are taken with $C' = 0^\circ, 45^\circ$, and 90° , and with $P = A = 0^\circ$. These three readings are, respectively,

$$\begin{aligned} R_4 &= (1 + s') g I_P, \\ R_5 &= (1 - f') g I_P, \\ R_6 &= (1 - s') g I_P. \end{aligned} \quad (20)$$

The defect parameters are seen to be

$$\begin{aligned} s &= (R_1 - R_3)/(R_1 + R_3), \\ f &= (R_1 - 2R_2 + R_3)/(R_1 + R_3), \\ s' &= (R_4 - R_6)/(R_4 + R_6), \\ f' &= (R_4 - 2R_5 + R_6)/(R_4 + R_6), \end{aligned} \quad (21)$$

and the detector response to the input intensity is given by

$$g I_P = (R_1 + R_3)/2 = (R_4 + R_6)/2. \quad (22)$$

This is a sufficient calibration of the ellipsometer, since r follows from s and f by Eqs. (4) and (6): $r = (1 - p^2 - s^2)^{1/2}$, where $p = 1 - 2f$. Similarly r' is found. The source intensity I_P need not be determined, unless absolute intensities (Stokes parameter values) are desired.²⁴ This is not required for MME, since $g I_P$ and $g I_A$ can be compared to infer $\mathbf{M}$. Once calibrated, and with the system in place, an appropriately selected set of readings, at least 16 in number, permits an absolute determination of the Mueller matrix $\mathbf{M}$. The remainder of the paper describes four such methods of MME.

III. DIRECT (16-INTENSITY) METHODS

Direct methods of MME involve recording 16 (or more) measurements of intensity to infer $\mathbf{M}$. It is common for linear and circular polarization "filters" (assumed to be ideal) to be used.^{25,26} A calibrated PCSC'A ellipsometer accounts for inevitable component imperfections and thus provides a more accurate measurement.

A. Varying $\mathbf{P}$, $\mathbf{C}$, $\mathbf{C}'$, and $\mathbf{A}$

For ideal compensators, the simplest method of MME is to adjust the azimuths P, C, C' , and A so that the PSG (PSD) generates (extinguishes) three linearly polarized states separated by 45° , and one circularly polarized state. This method is only slightly more complicated if the compensators are imperfect. Rotating the polarizer when there is residual (linear) source polarization, or rotating the analyzer when there is detector polarization sensitivity, introduces spurious variations in detector response. This is the only method described here for which $\mathbf{P}$ and $\mathbf{A}$ are not held fixed. The analysis properly accounts for compensator imperfections for the most straightforward 16-intensity method of MME.

The (P, C) azimuths of the PSG are set to the following values: $(0^\circ, 0^\circ)$, $(45^\circ, 45^\circ)$, $(90^\circ, 90^\circ)$, and $(0^\circ, 45^\circ)$. The resulting input Stokes parameter coefficients from Eq. (8) comprise the columns of the generator matrix $\mathbf{G}$:

$$\mathbf{G} = \begin{bmatrix} 1+s & 1+s & 1+s & 1 \\ 1+s & 0 & -1-s & 1-2f \\ 0 & 1+s & 0 & s \\ 0 & 0 & 0 & r \end{bmatrix}. \quad (23)$$

The (C', A) azimuths of the PSD are set to the following values: $(0^\circ, 0^\circ)$, $(45^\circ, 45^\circ)$, $(90^\circ, 90^\circ)$, and $(45^\circ, 0^\circ)$. The resulting Stokes parameter coefficients from Eq. (11) form the rows of the detector matrix $\mathbf{D}$:

$$\mathbf{D} = \begin{bmatrix} 1+s' & 1+s' & 0 & 0 \\ 1+s' & 0 & 1+s' & 0 \\ 1+s' & -1-s' & 0 & 0 \\ 1 & 1-2f' & s' & -r' \end{bmatrix}. \quad (24)$$

The measured response matrix for the 16 combinations of the above azimuth pairs, as derived in the previous section, is given by

$$\mathbf{R} = (gI_P/2)\mathbf{D}\mathbf{M}\mathbf{G}. \quad (17)$$

The columns of $\mathbf{R}$ correspond to the PSG azimuth settings; the elements of each column give the measured intensity for each PSD azimuth pair.

$\mathbf{M}$ is recovered from $\mathbf{R}$ by

$$\mathbf{M} = (2/gI_P)\mathbf{D}^{-1}\mathbf{R}\mathbf{G}^{-1}, \quad (18)$$

where

$$\mathbf{G}^{-1} = \frac{1/2}{1+s} \begin{bmatrix} 1 & 1 & -1 & (s-2+2f)/r \\ 0 & 0 & 2 & -2s/r \\ 1 & -1 & -1 & (s-2f)/r \\ 0 & 0 & 0 & (2+2s)/r \end{bmatrix} \quad (25)$$

and

$$\mathbf{D}^{-1} = \frac{1/2}{1+s'} \begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 0 & -1 & 0 \\ -1 & 2 & -1 & 0 \\ \frac{2-2f'-s'}{r'} & \frac{2s'}{r'} & \frac{2f'-s'}{r'} & \frac{-(2+2s')}{r'} \end{bmatrix} \quad (26)$$

B. Calibration

The compensator defect parameters and the response of the detector to the input intensity I_P may be found by the method of the last section, or by measuring the response matrix with the system absent. From Eqs. (23), (24), and (17) with $\mathbf{M}$ set equal to the identity matrix $\mathbf{I}$, the calibration response matrix $\mathbf{R}_C$ is given by

$$\mathbf{R}_C = (gI_P/2)\mathbf{D}\mathbf{I}\mathbf{G} = (gI_P/2) \begin{bmatrix} 2c & c & 0 & d \\ c & 2c & c & c \\ 0 & c & 2c & e \\ d' & c & e' & h \end{bmatrix}, \quad (27)$$

where

$$c \simeq d \simeq e \simeq d' \simeq e' \simeq 1, \quad (28)$$

$$h \simeq 0.$$

The approximations are strict equalities for ideal compensators. The parameters s, f, s' , and f' are obtained from

$$\begin{aligned} s &= 2c/(d+e) - 1, \\ f &= e/(d+e), \\ s' &= 2c/(d'+e') - 1, \\ f' &= e'/(d'+e'). \end{aligned} \quad (29)$$

Equations (29) may be evaluated from the elements of $\mathbf{R}_C$, since any multiplicative factor cancels out. The nine measurements related by c may be compared as a consistency check. The first calibration measurement $R_{C00} = gI_{PC}$ gives the source response in terms of the matrix element $c = (1+s)(1+s')$:

$$gI_P = R_{C00}/[(1+s)(1+s')]. \quad (30)$$

C. Varying only C and C'

When P and A are kept fixed throughout a measurement, the source may emit light of any constant intensity and polarization state (so long as there is a nonzero component along the polarizer transmitting axis). Correspondingly, the detector response may have any constant dependence on polarization state (so long as the response is not zero for the polarization produced by the analyzer). The following direct method fixes P and A, while rotating only C and C'.

Without loss of generality we fix $A = P = 0^\circ$. Equations (8) and (11) then become

$$\begin{bmatrix} S_0 \\ S_1 \\ S_2 \\ S_3 \end{bmatrix} = \begin{bmatrix} s \cos 2C + 1 \\ f \cos 4C + s \cos 2C + (1-f) \\ f \sin 4C + s \sin 2C \\ r \sin 2C \end{bmatrix} I_P \quad (31)$$

and

$$\begin{aligned} 2I_A &= S'_0[s' \cos 2C' + 1] \\ &\quad + S'_1[f' \cos 4C' + s' \cos 2C' + (1-f')] \\ &\quad + S'_2[f' \sin 4C' + s' \sin 2C'] \\ &\quad + S'_3[-r' \sin 2C']. \end{aligned} \quad (32)$$

We again determine the effect of the system on four input states (i.e., four values of C), from the intensities obtained for four PSD settings (i.e., four values of C'). The generator matrix, derived from Eq. (31), is thus written as

$$\mathbf{G} = \begin{bmatrix} 1 & s & 0 & 0 & 0 \\ 1-f & s & 0 & f & 0 \\ 0 & 0 & s & 0 & f \\ 0 & 0 & r & 0 & 0 \end{bmatrix} \times \begin{bmatrix} 1 & 1 & 1 & 1 \\ \cos 2C_1 & \cos 2C_2 & \cos 2C_3 & \cos 2C_4 \\ \sin 2C_1 & \sin 2C_2 & \sin 2C_3 & \sin 2C_4 \\ \cos 4C_1 & \cos 4C_2 & \cos 4C_3 & \cos 4C_4 \\ \sin 4C_1 & \sin 4C_2 & \sin 4C_3 & \sin 4C_4 \end{bmatrix} \quad (33)$$

and the detector matrix, derived from Eq. (32) becomes

$$\mathbf{D} = \begin{bmatrix} 1 \cos 2C'_1 & \sin 2C'_1 & \cos 4C'_1 & \sin 4C'_1 \\ 1 \cos 2C'_2 & \sin 2C'_2 & \cos 4C'_2 & \sin 4C'_2 \\ 1 \cos 2C'_3 & \sin 2C'_3 & \cos 4C'_3 & \sin 4C'_3 \\ 1 \cos 2C'_4 & \sin 2C'_4 & \cos 4C'_4 & \sin 4C'_4 \end{bmatrix} \times \begin{bmatrix} 1 & 1-f' & 0 & 0 \\ s' & s' & 0 & 0 \\ 0 & 0 & s' & -r' \\ 0 & f' & 0 & 0 \\ 0 & 0 & f' & 0 \end{bmatrix} \quad (34)$$

$\mathbf{R}$ is obtained by substituting Eqs. (33) and (34) into

$$\mathbf{R} = (gI_P/2)\mathbf{D}\mathbf{M}\mathbf{G}. \quad (17)$$

If we choose azimuth values of (0° , 22.5° , 45° , and 67.5°) for both C and C' then $\mathbf{G}$ and $\mathbf{D}$ possess inverses given by

$$\mathbf{G}^{-1} = \frac{1}{2Df} \times \begin{bmatrix} \frac{2(Df - E)}{H} & \frac{2E}{H} & \frac{-2BDs}{H} & \frac{2(2BDs^2 - f)}{Hr} \\ -1 & 1 & D & \frac{(2f - Ds)}{r} \\ 2B & -2B & 0 & \frac{-2f}{r} \\ -1 & 1 & -D & \frac{(2f + Ds)}{r} \end{bmatrix} \quad (35)$$

and

$$\mathbf{D}^{-1} = \frac{1}{2Df'} \times \begin{bmatrix} \frac{2(Df' - E)}{H'} & -1 & 2B & -1 \\ \frac{2E}{H'} & 1 & -2B & 1 \\ \frac{-2BDs'}{H'} & D & 0 & -D \\ \frac{2(f' - 2BDs'^2)}{H'r'} & \frac{(Ds' - 2f')}{r'} & \frac{2f'}{r'} & \frac{-(2f' + Ds')}{r'} \end{bmatrix} \quad (36)$$

where

$$B = 2^{-1/2}, \quad D = 2B - 1, \quad E = B - 1, \\ H = 1 + s, \quad H' = 1 + s'. \quad (37)$$

$\mathbf{M}$ is recovered from $\mathbf{R}$ by substituting Eqs. (35)–(37) into

$$\mathbf{M} = (2/gI_P)\mathbf{D}^{-1}\mathbf{R}\mathbf{G}^{-1}. \quad (18)$$

D. Calibration

The compensator defect parameters may be determined by a partial determination of the response matrix made with the system absent. From Eqs. (17), (33), and (34) with $\mathbf{M}$ set equal to $\mathbf{I}$, the calibration response matrix $\mathbf{R}_C$ is given by

$$\mathbf{R}_C = (gI_P/2)\mathbf{DIG} = (gI_P/2) \begin{bmatrix} c & d & e & h \\ d' & - & - & - \\ e' & - & - & - \\ h' & - & - & - \end{bmatrix}, \quad (38)$$

where

$$c \simeq 1, \quad d \simeq h \simeq d' \simeq h' \simeq 0.75, \quad e \simeq e' \simeq 0.5. \quad (39)$$

The nine missing calibration measurements need not be taken. The parameters s , f , s' , and f' are obtained from

$$\begin{aligned} s &= (d - h)/[c\sqrt{2} - (d - h)], \\ f &= [2e - (d + h)]/[e - (d + h)], \\ s' &= (d' - h')/[c\sqrt{2} - (d' - h')], \\ f' &= [2e' - (d' + h')]/[e' - (d' + h')]. \end{aligned} \quad (40)$$

Equations (40) may be evaluated from the elements of $\mathbf{R}_C$. The first calibration measurement $R_{C00} = gI_{PC}/2$ again gives

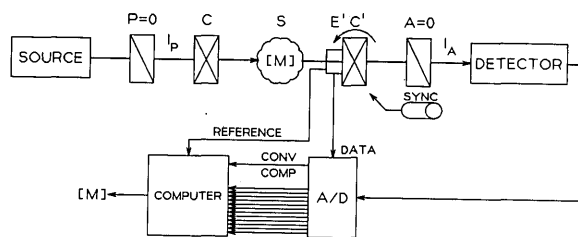


FIG. 3. Semi-automatic (rotating-compensator) Mueller matrix ellipsometer. An encoder E' mounted on C' provides pulses for digitizing the detector response, and a reference pulse to synchronize data acquisition. C is stepped after n rotations of C' , either manually or by a stepping motor under computer control (not shown).

the source response in terms of the matrix element $c = 2(1 + s)(1 + s')$:

$$gI_P = R_{C00}/[(1 + s)(1 + s')]. \quad (30)$$

IV. SEMIAUTOMATIC (ROTATING-COMPENSATOR) METHOD

Direct methods, when performed manually, are slow and subject to azimuth positioning errors. Programmed stepping motors could be used to position the components, along with automatic data recording. However, since 16 matrix elements are deduced from 16 intensity measurements, no determination of experimental error is available. (An exception to this occurs when certain matrix elements are known *a priori* to be equal or zero.) Two automated methods are now described, by which the Mueller matrix is overdetermined. The methods are significantly faster and more precise, and provide additional consistency checks to monitor errors. In the following method, the PSD is automated.

A. Automation of PSD

Figure 3 shows a schematic representation of the method. The input arm of the ellipsometer is unchanged. In the output arm, a Fourier-transform detection technique previously developed for rotating-analyzer ellipsometers is used.^{27,28} The compensator C' is rotated continuously in front of the fixed analyzer A by a synchronous motor, and monitored by a shaft encoder E' .²⁹ The encoder provides a single reference pulse each rotation as one output, and data pulses at m equal angular increments per rotation as another output. The analog output of the detector is digitized on signal from the data pulses, and read sequentially into a small computer on signal from the conversion-complete pulse of the A/D converter. The entire process is synchronized by having the reading inhibited until enabled by the encoder's reference pulse. A fixed number of data, corresponding to an integral number n of compensator rotations, are read under program control. Accumulating (averaging) the data for each value of C' reduces the variance of the readings by a factor n .

The detector response, according to Eqs. (11) and (14), can be expressed as a Fourier series in C' :

$$R(C') = A_0 + A_2 \cos 2C' + B_2 \sin 2C' \\ + A_4 \cos 4C' + B_4 \sin 4C', \quad (41)$$

where, setting $A = 0^\circ$,

$$\begin{aligned}
A_0 &= (g/2)[S'_0 + (1-f')S'_1], \\
A_2 &= (g/2)[s'S'_0 + s'S'_1], \\
B_2 &= (g/2)[s'S'_2 - r'S'_3], \\
A_4 &= (g/2)[f'S'_1], \\
B_4 &= (g/2)[f'S'_2].
\end{aligned} \quad (42)$$

The five coefficients in Eqs. (42) contain all the polarimetric information available from the measurement. Digital Fourier analysis gives the best least-squares representation of the raw data and excludes much of the random and systematic error. Other harmonics are present only due to noise or misalignment. Their (quadrature-summed) magnitude gives an indication of measurement quality which is independent of the polarimetric content of the data.

Solving Eqs. (42) for the output Stokes parameters relative to I_P , we obtain

$$\begin{aligned}
S'_0 &= (2/gI_P)[A_0 - A_4(1-f')/f']I_P, \\
S'_1 &= (2/gI_P)[A_4/f']I_P, \\
S'_2 &= (2/gI_P)[B_4/f']I_P, \\
S'_3 &= (2/gI_P)[-B_2 + B_4(s'/f')]I_P/r'.
\end{aligned} \quad (43)$$

Thus, a complete determination of polarization state is obtained from Fourier analysis of a single waveform.²⁴

The first description of this (rotating-compensator, fixed-analyzer) PSD as a polarimeter was given by Sekera,³⁰ who used analog methods to extract the harmonically coded information. Hodgdon³¹ described its calibration for an imperfect compensator and partial polarizer, and used digital multiple linear regression to decode the information. Orrall³² described its use for a retarder having arbitrary δ_C (but $\psi_C = 45^\circ$), and used digital Fourier analysis with a minicomputer to obtain the coefficients in Eqs. (42).

The Sekera polarimeter was first used in ellipsometry by the present author³³ to determine the complete system Jones matrix (generalized ellipsometry) with an imperfect compensator³⁴ and for MME, assuming an ideal compensator.³⁵ The present analysis removes that assumption for MME. Other forms of rotating-retarder PSD's have been described for ellipsometry of depolarizing systems.^{10,36,37} Reinvention of the Sekera Fourier polarimeter has been accomplished by other workers.³⁸

B. Analysis

To perform MME with imperfect compensators, the polarizer and analyzer are fixed at $P = A = 0^\circ$. The input Stokes vector varies with C [rewriting Eq. (31)] according to

$$\begin{bmatrix} S_0 \\ S_1 \\ S_2 \\ S_3 \end{bmatrix} = \begin{bmatrix} 1 & s & 0 & 0 & 0 \\ 1-f & s & 0 & f & 0 \\ 0 & 0 & s & 0 & f \\ 0 & 0 & r & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ \cos 2C \\ \sin 2C \\ \cos 4C \\ \sin 4C \end{bmatrix} I_P. \quad (44)$$

From Eq. (9), the output Stokes vector is

$$\begin{aligned}
\begin{bmatrix} S'_0 \\ S'_1 \\ S'_2 \\ S'_3 \end{bmatrix} &= \begin{bmatrix} M_{00} & M_{01} & M_{02} & M_{03} \\ M_{10} & M_{11} & M_{12} & M_{13} \\ M_{20} & M_{21} & M_{22} & M_{23} \\ M_{30} & M_{31} & M_{32} & M_{33} \end{bmatrix} \\
&\times \begin{bmatrix} 1 & s & 0 & 0 & 0 \\ 1-f & s & 0 & f & 0 \\ 0 & 0 & s & 0 & f \\ 0 & 0 & r & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ \cos 2C \\ \sin 2C \\ \cos 4C \\ \sin 4C \end{bmatrix} I_P \quad (45) \\
&= \begin{bmatrix} E_{00} & E_{02} & F_{02} & E_{04} & F_{04} \\ E_{10} & E_{12} & F_{12} & E_{14} & F_{14} \\ E_{20} & E_{22} & F_{22} & E_{24} & F_{24} \\ E_{30} & E_{32} & F_{32} & E_{34} & F_{34} \end{bmatrix} \begin{bmatrix} 1 \\ \cos 2C \\ \sin 2C \\ \cos 4C \\ \sin 4C \end{bmatrix} I_P. \quad (46)
\end{aligned}$$

Thus, each of the output Stokes parameters S'_k ($k = 0, \dots, 3$) can be expressed as a Fourier series in C :

$$S'_k = (E_{k0} + E_{k2} \cos 2C + F_{k2} \sin 2C + E_{k4} \cos 4C + F_{k4} \sin 4C) I_P. \quad (47)$$

The output Stokes parameters S'_k are measured relative to I_P for a number l (e.g., $l = 8$ or 12) of compensator settings C_q , beginning at 0° (i.e., $C = P$) with equal increments over a 180° range:

$$C_q = (q-1) 180^\circ/l, \quad (q = 1, \dots, l). \quad (48)$$

The S'_k values are then subjected to Fourier analysis with respect to C to determine the matrix elements E_{kj} and F_{kj} in Eqs. (46) and (47).

If we call the matrices to the right of the equals sign in Eqs. (44) and (46) $\mathbf{B}$ and $\mathbf{E}$, respectively, we have simply

$$\mathbf{E} = \mathbf{M}\mathbf{B}. \quad (49)$$

The elements of $\mathbf{E}$ are determined from the second Fourier analysis, while the elements of $\mathbf{B}$ are determined from the calibration of C .

The significance of Eq. (49) lies in the fact that lmn measured intensities are condensed by means of two separate Fourier analyses into the 20 elements of $\mathbf{E}$, no matter how large l , m , and n are chosen to be. Thus, the measurement may be greatly overdetermined without increasing the complexity of the solution for $\mathbf{M}$.³⁹ The least-squares solution of Eq. (49) is

$$\mathbf{M} = \mathbf{E}\mathbf{B}^t(\mathbf{B}\mathbf{B}^t)^{-1} = \mathbf{E}\mathbf{F}, \quad (50)$$

where $\mathbf{F} = \mathbf{B}^t(\mathbf{B}\mathbf{B}^t)^{-1}$ depends only on the defect parameters of the compensator C . From Eq. (44), $\mathbf{F}$ is given by

$$\mathbf{F} = \begin{bmatrix} (f+s^2)/h & -s^2/h & 0 & 0 \\ s(2f-1)/h & s/g & 0 & 0 \\ 0 & 0 & 0 & 1/r \\ (f-1-s^2)/h & (1+s^2)/h & 0 & 0 \\ 0 & 0 & 1/f & -s/rf \end{bmatrix}, \quad (51)$$

where

$$h = f(2s^2 + 1) \simeq 0.5. \quad (52)$$

This result can be verified by noting that from Eqs. (49) and (50), $\mathbf{M} = \mathbf{M}\mathbf{B}\mathbf{F}$ and that from Eqs. (44), (51), and (52) $\mathbf{B}\mathbf{F}$ gives the identity matrix.

C. Calibration

The compensator azimuths C' at which readings are taken are given by

$$C'_k = C'_1 + (k - 1)360^\circ/m, \quad (k = 1, \dots, m), \quad (53)$$

where C'_1 is the first-datum azimuth. The sampled detector response is thus a Fourier series in the variable $(C' - C'_1)$:

$$R(C' - C'_1) = A_0' + A_2' \cos 2(C' - C'_1) + B_2' \sin 2(C' - C'_1) + A_4' \cos 4(C' - C'_1) + B_4' \sin 4(C' - C'_1). \quad (54)$$

A limited Fourier analysis of this data^{27,28} gives the primed coefficients in Eq. (54). This is a high-precision measurement, since five coefficient values are deduced from n repeated measurements of R at m different values of C' . The unprimed coefficients needed in Eqs. (43) to obtain the output Stokes parameters are then given by

$$\begin{aligned} A_0 &= A_0', \\ A_2 &= [A_2' \cos 2C'_1 - B_2' \sin 2C'_1], \\ B_2 &= [B_2' \cos 2C'_1 + A_2' \sin 2C'_1], \\ A_4 &= [A_4' \cos 4C'_1 - B_4' \sin 4C'_1], \\ B_4 &= [B_4' \cos 4C'_1 + A_4' \sin 4C'_1]. \end{aligned} \quad (55)$$

To determine C'_1 , as well as the compensator defect parameters, the arms are made collinear with no system present ($S' = S$) and $P = C = A = 0^\circ$. From Eqs. (8) and (42), $S_2 = S_3 = 0$ and $S_1 = S_0$. Thus,

$$\begin{aligned} A_0 &= (g/2)(2 - f')S_0, \\ A_2 &= gs'S_0, \\ B_2 &= 0, \\ A_4 &= (g/2)f'S_0, \\ B_4 &= 0. \end{aligned} \quad (56)$$

Combining the last of Eqs. (55) and (56), we obtain

$$\tan 4C'_1 = -B_4'/A_4'. \quad (57)$$

The compensator C' may be adjusted in its holder until $C'_1 = 0^\circ$ (i.e., the measured value of B_4' is zero). Alternatively, C' may be left arbitrarily positioned. In this case, C'_1 is determined by Eq. (57) for use in Eqs. (55). In either case, the angular offset of the second compensator is properly accounted for, and the unprimed coefficients are found. From Eqs. (56), the defect parameters are given by

$$\begin{aligned} s' &= A_2/(A_0 + A_4), \\ f' &= 2A_4/(A_0 + A_4). \end{aligned} \quad (58)$$

The defect parameters for C follow from a relative measurement (gI_P is not yet known) of $\mathbf{E} = \mathbf{I}\mathbf{B} = \mathbf{B}$ with the system removed. The normalized matrix elements $e_{ij} = E_{ij}/E_{00}$ and $f_{ij} = F_{ij}/E_{00}$, are independent of gI_P , so that from Eqs. (45) and (46)

$$\begin{aligned} s &= e_{02} = e_{12} = f_{22}, \\ f &= e_{14} = f_{24} = 1 - e_{10}. \end{aligned} \quad (59)$$

The equalities above serve as a consistency check on the calibration. The source response is determined by equating the expression for S'_0 from Eqs. (43) with that for S_0 from Eq. (44):

$$(2/g)[A_0 - A_4(1 - f')/f'] = (1 + s \cos 2C)I_P. \quad (60)$$

This gives a value of gI_P for each azimuth C . A better determination is obtained by averaging the coefficients for all C values. Denoting average over C by $\langle \rangle$, Eq. (60) becomes

$$gI_P = 2[\langle A_0 \rangle - \langle A_4 \rangle(1 - f')/f']. \quad (61)$$

D. Review of the method

Once the ellipsometer is calibrated, $\mathbf{M}$ is determined as follows: The system is inserted, the arms are positioned at the desired scattering angle and the compensator C is set at $C = 0^\circ$. The primed coefficients in Eq. (54) obtained from Fourier analysis of the raw data are transformed by Eqs. (55) to give the unprimed coefficients. Eqs. (43) give the relative output Stokes parameters. The procedure is repeated for the l values of C given by Eq. (48). The l sets of output Stokes parameters are Fourier analyzed to obtain the elements of the matrix $\mathbf{E}$ in Eq. (46). $\mathbf{M}$ then follows from Eq. (50).

The computer which performs the two Fourier analyses is programmed for the other calculations as well. The operator need only position C to its l different azimuths and start the measurement each time. Alternatively, a stepping motor could perform this function under program control, making this method fully automatic.

V. AUTOMATIC (DUAL ROTATING-COMPENSATOR) METHOD

In the previous method, by rotating C' , all four output Stokes parameters were determined from the Fourier analysis of a single waveform. In this method, the compensators are rotated simultaneously, and all 16 Mueller matrix elements are determined from the Fourier analysis of a more complicated waveform. This method has been described for perfect compensators and perfect alignment by Azzam.⁴⁰

A. Analysis

To begin, Eqs. (6), (7), (9), (11), and (14) are combined to obtain the detector output as

$$R(C, C') = (gI_P/2) \left\{ a_0 + \sum_{j=1}^{12} a_j \cos \theta_j + b_j \sin \theta_j \right\}, \quad (62)$$

where the angles θ_j are given by

$$\begin{aligned} \theta_1 &= 2C, \\ \theta_2 &= 4C, \\ \theta_3 &= 2C' - 4C, \\ \theta_4 &= 2C' - 2C, \\ \theta_5 &= 2C', \\ \theta_6 &= 2C' + 2C, \\ \theta_7 &= 2C' + 4C, \\ \theta_8 &= 4C' - 4C, \\ \theta_9 &= 4C' - 2C, \\ \theta_{10} &= 4C', \\ \theta_{11} &= 4C' + 2C, \\ \theta_{12} &= 4C' + 4C. \end{aligned} \quad (63)$$

The coefficients, setting $P = A = 0^\circ$, depend on $\mathbf{M}$ and the compensator defect parameters according to

$$\begin{aligned}
a_0 &= M_{00} + (1-f)M_{01} + (1-f')M_{10} + (1-f)(1-f')M_{11}, \\
a_1 &= s[(M_{00} + M_{01}) + (1-f')(M_{10} + M_{11})], \\
b_1 &= s[M_{02} + (1-f')M_{12}] + r[M_{03} + (1-f')M_{13}], \\
a_2 &= f[M_{01} + (1-f')M_{11}], \\
b_2 &= f[M_{02} + (1-f')M_{12}], \\
a_3 &= f[s'(M_{01} + M_{11} + M_{22}) - r'M_{32}]/2, \\
b_3 &= -f[s'(M_{02} + M_{12} - M_{21}) + r'M_{31}]/2, \\
a_4 &= [ss'(M_{00} + M_{01} + M_{10} + M_{11} + M_{22}) \\
&\quad + rs'M_{23} - r'sM_{32} - rr'M_{33}]/2, \\
b_4 &= [ss'(M_{20} + M_{21} - M_{12} - M_{02}) \\
&\quad - rs'(M_{13} + M_{03}) - r's(M_{30} + M_{31})]/2, \\
a_5 &= s'[(M_{00} + M_{10}) + (1-f)(M_{01} + M_{11})], \\
b_5 &= s'[M_{20} + (1-f)M_{21}] - r'[M_{30} + (1-f)M_{31}], \\
a_6 &= [ss'(M_{00} + M_{01} + M_{10} + M_{11} - M_{22}) \\
&\quad - rs'M_{23} + r'sM_{32} + rr'M_{33}]/2, \quad (64) \\
b_6 &= [ss'(M_{20} + M_{21} + M_{12} + M_{02}) \\
&\quad + rs'(M_{13} + M_{03}) - r's(M_{30} + M_{31})]/2, \\
a_7 &= f[s'(M_{01} + M_{11} - M_{22}) + r'M_{32}]/2, \\
b_7 &= f[s'(M_{02} + M_{12} + M_{21}) - r'M_{31}]/2, \\
a_8 &= ff'(M_{11} + M_{22})/2, \\
b_8 &= ff'(M_{21} - M_{12})/2, \\
a_9 &= f'[s(M_{10} + M_{11} + M_{22}) + rM_{23}]/2, \\
b_9 &= f'[s(M_{20} - M_{12} + M_{21}) - rM_{13}]/2, \\
a_{10} &= f'[M_{10} + (1-f)M_{11}], \\
b_{10} &= f'[M_{20} + (1-f)M_{21}], \\
a_{11} &= f'[s(M_{10} + M_{11} - M_{22}) - rM_{23}]/2, \\
b_{11} &= f'[s(M_{20} + M_{12} + M_{21}) + rM_{13}]/2, \\
a_{12} &= ff'(M_{11} - M_{22})/2, \\
b_{12} &= ff'(M_{21} + M_{12})/2.
\end{aligned}$$

From these 25 coefficients, the 16 matrix elements can be recovered:

$$\begin{aligned}
M_{00} &= a_0 - a_2(1-f)/f - a_{10}(1-f')/f' \\
&\quad + (a_8 + a_{12})(1-f)(1-f')/ff',
\end{aligned}$$

$$\begin{aligned}
M_{11} &= (a_8 + a_{12})/ff', \\
M_{22} &= (a_8 - a_{12})/ff', \\
M_{12} &= -(b_8 - b_{12})/ff', \\
M_{21} &= (b_8 + b_{12})/ff', \\
M_{01} &= [a_2 - (a_8 + a_{12})(1-f')/f]/f, \\
M_{02} &= [b_2 + (b_8 - b_{12})(1-f')/f]/f, \\
M_{10} &= [a_{10} - (a_8 + a_{12})(1-f)/f]/f', \\
M_{20} &= [b_{10} - (b_8 + b_{12})(1-f)/f]/f', \\
M_{13} &= [(b_{11} - b_9)/f' - sM_{12}]/r, \\
M_{23} &= [(a_9 - a_{11})/f' - sM_{22}]/r, \\
M_{32} &= [(a_7 - a_3)/f + s'M_{22}]/r', \\
M_{31} &= [-(b_7 + b_3)/f + s'M_{21}]/r', \\
M_{30} &= [-b_5 + (b_7 + b_3)(1-f)/f + s'M_{20}]/r', \\
M_{03} &= [b_1 + (b_9 - b_{11})(1-f')/f' - sM_{02}]/r, \\
M_{33} &= [(a_6 - a_4) + s'rM_{23} - sr'M_{32} + ss'M_{22}]/rr',
\end{aligned} \quad (65)$$

as well as 9 other relationships which serve as consistency checks:

$$\begin{aligned}
a_1 &\simeq b_4 \simeq a_5 \simeq b_6 \simeq 0, \\
a_7 + a_3 &= fs'(M_{01} + M_{11}) \simeq 0, \\
b_7 - b_3 &= fs'(M_{02} + M_{12}) \simeq 0, \\
a_9 + a_{11} &= f's(M_{10} + M_{11}) \simeq 0, \\
b_9 + b_{11} &= f's(M_{20} + M_{21}) \simeq 0, \\
a_6 + a_4 &= ss'(M_{00} + M_{01} + M_{10} + M_{11}) \simeq 0.
\end{aligned} \quad (66)$$

To transform Eq. (62) into a Fourier series, C and C' are rotated synchronously in a relationship which makes the angles θ_j distinct and harmonically related. Optimally, θ_j should be as closely spaced as possible. A rotation ratio of 5:1 fulfills these conditions.⁴¹ That is, if

$$C' - C_1 = 5(C - C_1), \quad (67)$$

where $C = C_1$ and $C' = C'_1$ at the first datum, then

$$\theta_j = 2j(C - C_1) + \varphi_j, \quad (j = 1, \dots, 12). \quad (68)$$

The phase angle corrections are given by

$$\begin{aligned}
\varphi_1 &= 2C_1, \\
\varphi_2 &= 4C_1, \\
\varphi_3 &= 2C'_1 - 4C_1, \\
\varphi_4 &= 2C'_1 - 2C_1, \\
\varphi_5 &= 2C'_1, \\
\varphi_6 &= 2C'_1 + 2C_1, \\
\varphi_7 &= 2C'_1 + 4C_1, \\
\varphi_8 &= 4C'_1 - 4C_1, \\
\varphi_9 &= 4C'_1 - 2C_1, \\
\varphi_{10} &= 4C'_1, \\
\varphi_{11} &= 4C'_1 + 2C_1, \\
\varphi_{12} &= 4C'_1 + 4C_1.
\end{aligned} \quad (69)$$

The detector response becomes a Fourier series in the single variable $2(C - C_1)$:

$$\begin{aligned}
R(C - C') &= A'_0 + \sum_{j=1}^{12} A'_j \cos 2j(C - C') \\
&\quad + B'_j \sin 2j(C - C'). \quad (70)
\end{aligned}$$

By recording the detector response m times per rotation of C', at azimuths of

$$C'_k = C'_1 + (k - 1)360^\circ/m, \quad (k = 1, \dots, m), \quad (53)$$

as in the previous method, a complete waveform sample, consisting of $5m$ detector readings, is obtained. This corresponds to a single rotation of C, or five rotations of C'. Fourier analysis^{27,28} with respect to $2(C - C_1)$ gives the primed coefficients in Eq. (70). The unprimed coefficients in Eq. (62) are recovered by the following transformation:

$$\begin{aligned}
a_0 &= (2/gI_P)A'_0, \\
a_j &= (2/gI_P)[A'_j \cos \varphi_j - B'_j \sin \varphi_j], \quad (j = 1, \dots, 12), \\
b_j &= (2/gI_P)[B'_j \cos \varphi_j + A'_j \sin \varphi_j].
\end{aligned} \quad (71)$$

Equations (65) then give the elements of M.

B. Calibration

Calibration consists of a determination of the defect parameters, source response gI_P and phase corrections φ_j . The latter derive from the first-datum compensator azimuths (C_1 , C'_1) by Eqs. (69).

A measurement is performed with the system absent. The unprimed coefficients, substituting $\mathbf{M} = \mathbf{I}$ in Eqs. (64), are

$$\begin{aligned}
 a_0 &= 1 + (1-f)(1-f') \simeq 1.25, \\
 a_1 &= s(2-f') \simeq 0, \\
 a_2 &= f(1-f') \simeq 0.25, \\
 a_3 &= fs' \simeq 0, \\
 a_4 &= (3ss' - rr')/2 \simeq -0.5, \\
 a_5 &= s'(2-f) \simeq 0, \\
 a_6 &= (ss' + rr')/2 \simeq 0.5, \\
 a_7 &= 0, \\
 a_8 &= ff' \simeq 0.25, \\
 a_9 &= sf' \simeq 0, \\
 a_{10} &= f'(1-f) \simeq 0.25, \\
 a_{11} &= 0, \\
 a_{12} &= 0,
 \end{aligned} \tag{72}$$

and

$$b_j = 0, \quad (j = 1, \dots, 12). \tag{73}$$

Since all the b_j are zero, the last of Eqs. (71) gives the phase corrections as

$$\varphi_j = -\tan^{-1}(B'_j/A'_j), \quad (j = 1, \dots, 12). \tag{74}$$

From Eqs. (72) it is clear that in the absence of a system, only the even coefficients up to the 10th have appreciable magnitude, making only those φ_j reliably measurable. Since the harmonics having the largest magnitude are the 4th and 6th, we find using Eqs. (69)

$$\begin{aligned}
 C_1 &= (\varphi_6 - \varphi_4)/4, \\
 C'_1 &= (\varphi_6 + \varphi_4)/4,
 \end{aligned} \tag{75}$$

and obtain three consistency checks:

$$\begin{aligned}
 \varphi_4 &= (\varphi_{10} - \varphi_2)/2, \\
 \varphi_6 &= (\varphi_{10} + \varphi_2)/2, \\
 \varphi_8 &= (\varphi_{10} - \varphi_2).
 \end{aligned} \tag{76}$$

From Eqs. (69) and (75) the unmeasurable φ_j are calculated. Equations (72) and (77) and the corrected cosine coefficients

$$A_j = (gI_P/2)a_j = [A'_j \cos\varphi_j - B'_j \sin\varphi_j], \quad (j = 1, \dots, 12), \tag{77}$$

give the source response and defect parameters as

$$gI_P = 2(A_8 + A_{10})(A_8 + A_2)/A_8 \tag{78}$$

and

$$\begin{aligned}
 f &= A_8/(A_8 + A_{10}), \\
 f' &= A_8/(A_8 + A_2), \\
 s &= (A_1 + A_9)/gI_P, \\
 s' &= (A_3 + A_5)/gI_P.
 \end{aligned} \tag{79}$$

C. Implementation

The compensators may be rotated to satisfy Eq. (67) by a number of means. A direct mechanical link (gears, pulleys) could be used. Since C and C' are mounted on opposite arms of the instrument, this approach may pose mechanical difficulties when making changes in the scattering angle. Sep-

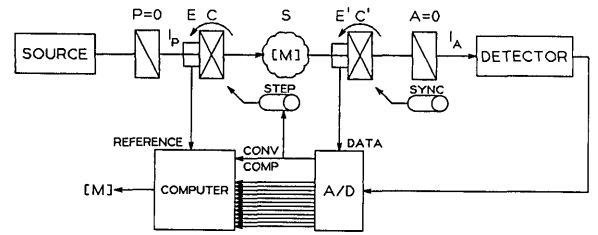


FIG. 4. Automatic (dual rotating-compensator) Mueller matrix ellipsometer. Encoder E' provides data pulses, while encoder E (on C) signals the beginning and ending of a complete optical cycle. C is stepped after each A/D conversion by one-fifth of the increment of E'.

arate synchronous motors driven from the same source would avoid this difficulty. Similarly, stepping motors could be used, with one driven by every 5th pulse of the other.

A fourth possibility, shown in Fig. 4, has some similarity to the previous configuration. C' is rotated by a synchronous motor and monitored by an angular encoder E', which supplies m equally spaced pulses per rotation. These pulses signal an A/D converter to sample and digitize the detector output. The conversion-complete pulse from the A/D converter signals the computer to read the digitized output. It also drives a stepping motor to advance C by one fifth of the encoder E' spacing. This process is completed before the next pulse from E' occurs. An encoder E monitors the position of C and puts out a single pulse per rotation. This pulse signals the beginning and end of a full measurement cycle (five rotations of C'). A total of $5m$ readings are taken, which constitute a sampled representation of Eq. (70). As before, multiples of $5m$ readings can be accumulated to increase precision.

VI. SUMMARY

The operation of the PCSC/A ellipsometer to perform MME in four different modes has been described. Although in theory all calibration methods are equivalent, there is a practical convenience (if not a gain in accuracy) in performing the calibration and measurement by similar procedures. Therefore, calibration procedures closely related to each method are given. These determine the azimuth, retardation and transmission ratio of both compensators so that spectroscopic measurements using achromatic retarders where all the above parameters are slightly wavelength dependent may be made. Calibration need not be repeated at a given wavelength.

Matrices have been used extensively in this paper, partly for notational compactness. It should not be concluded that the calculations they represent are overly complicated or time consuming. A digital computer has been assumed for the last two methods, and may be used in connection with the first two also. All the matrix inversions involved in the analysis have been performed analytically, so that none need be calculated. Thus, for each method, both the calibration and the measurement of $\mathbf{M}$ involve (beside the Fourier analyses in the last two methods) only straightforward evaluation of formulas.

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